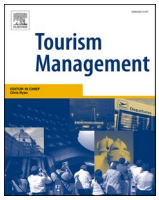
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Unlocking the power of multimodal online reviews: A multisensory perspective

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ABSTRACT

Multimodal online reviews have been an essential marketing tool for dominating consumer perception. However, the precise impact of visual, auditory, tactile, gustatory, and olfactory cues in multimodal content on consumers' perceived review usefulness remains unclear. This study aims to fill this gap by exploring how text and image contents affect review usefulness from a multisensory perspective. To this end, we proposed a framework that combined machine learning and multimodal large language models to quantify the multisensory elements conveyed through review text and images. Based on the Poisson regression model, the estimation results show that while multisensory elements in text and images have a generally consistent impact on consumer perception, each sensory element carries distinct value. Furthermore, the sensory discrepancy in text and images has a negative effect on the perceived review usefulness. Our findings contribute theoretically to enhancing review persuasion effect and sensory marketing literature.

1. Introduction

In recent years, the role of multimodal online reviews has gained increasing attention as a powerful force in shaping consumer behavior and influencing online interactions (Lee & Park, 2023; Li et al., 2023a, 2024a; Mahrous & Hassan, 2017; Plank, 2016). Therefore, a comprehensive analytical approach to multimodal online review is essential. Traditional analyses often rely on quantitative and qualitative methods but frequently overlook the sensory elements in multimodal online reviews (Gibson, 1979). As social media and online forums have evolved, individuals now express their opinions and experiences through various formats such as text, images, and videos (Hou & Pan, 2023; Li et al., 2022, 2023b; Zhu et al., 2025). However, the influence of these elements has not been adequately addressed in the existing literature (Fong et al., 2023). For example, the choice of words, tone, and punctuation in an online review can affect potential customers' perceptions and initial attitudes (Lee & Park, 2023; Li, Kwok, et al., 2023). Similarly, the colors, composition, and visual effects in a review image can directly influence their perceptions of the product or service (Li, Zhang, & Hsu, 2023). These impacts of multimodal review content primarily come from consumers' perception of specific content through different sensory channels (Krishna, 2012; Schmitt, 1999). Building on sensory marketing theory, which emphasizes the role of sensory cues—including visual, auditory, tactile, gustatory, and olfactory elements—in influencing consumer perceptions and behavior, this study proposes a research framework that extends this theory to the context of multimodal online reviews. By integrating the sensory elements of text and images, as well as their discrepancies, this framework aims to provide a more comprehensive understanding of how multisensory cues influence consumers' perceptions of review usefulness in the digital context.

In the context of online reviews, the concept of review usefulness has emerged as a fundamental metric for evaluating the value of multimodal contents, playing a pivotal role in shaping potential consumers' purchasing decisions (Barreda et al., 2020; Wang et al., 2023; Zhang et al., 2023). In the tourism and hospitality industry, useful reviews offer relevant information on facilities, services, and real experiences,

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providing strong support for potential consumers' decision-making processes (Ali & Ahmed, 2019; Li et al., 2022, 2023a; Zhang & Luo, 2023). Consequently, the study of review usefulness has become a prominent topic in hospitality research (Li, Zhang, & Hsu, 2023). A growing body of research indicates that review usefulness is influenced not only by content but also by presentation style, emotional tone, and the expression of subjective opinions (Li et al., 2023b, 2024a; Wang et al., 2024; Zhang et al., 2023). Existing studies have confirmed that elements like the reviewer's credibility and content relevance significantly impact review usefulness (Li, Li, et al., 2024; Wang et al., 2023; Zhang et al., 2023), but multisensory elements have been largely overlooked. For example, descriptions of food flavors and bed texture in text reviews can directly influence potential consumers' perceptions and evaluations, yet these sensory elements are often ignored or difficult to quantify (Alyahya & McLean, 2022; Lee et al., 2010). Similarly, in image reviews, the choice and presentation of visual elements can significantly influence review usefulness (Li et al., 2023a, 2024a). Compared to text, images more directly trigger potential consumers' sensory experiences (Li, Li et al., 2024b; Mehraliyev et al., 2020). This advantage lies in the intuitive nature of content and the swiftness of information processing. Studies have demonstrated that the human brain processes images 60,000 times faster than text (Baddeley, 2012). However, current research on sensory cues has primarily focused on exploring text reviews through manual evaluation. Therefore, research on "sensory experience" has limitations, especially in examining how the multisensory elements conveyed through text and images influence consumers' perceived value. Addressing this issue is likely to attract significant interest from researchers and practitioners in the fields of tourism multimodal data and sensory marketing. This serves as the primary motivation for this study.

In addition, according to sensory marketing theory (Krishna, 2012; Schmitt, 1999), the multisensory elements in review text and images are not the only factors influencing consumers' perceived value or experience; another key factor—multisensory discrepancies—stimulates consumers' sensory experiences. These discrepancies can enhance review effectiveness through complementarity but also pose risks of negative outcomes. In tourism and hospitality, such discrepancies are pronounced. Images, with their vividness and immediacy, convey sensory qualities like the perceived accommodation texture but may create overly idealized impressions that differ from textual descriptions, leading to confusion or unmet expectations (Li, Li, et al., 2024b; Mehraliyev et al., 2020). Text, by providing detailed narratives and personal evaluations, can either resolve or exacerbate these inconsistencies. Such divergences, while potentially enriching consumer understanding, may also undermine trust when misalignment arises, emphasizing the dual-edged nature of these sensory discrepancies (Li, Kwok, et al., 2023; Zhang et al., 2023). Therefore, effectively analyzing both can significantly enhance the informational value of multimodal review content and the depth of sensory experience, ultimately influencing consumers' decision-making processes. However, despite their significance, the impact of multisensory elements and their discrepancies in image and text information within multimodal online reviews on the value of review content remains unclear.

To address the above research gaps, this study aims to explore the value effects of text and images in multimodal online reviews from a multisensory perspective, focusing on their role in perceived reviews usefulness. Specifically, the study has three objectives: 1) to investigate how multisensory elements in textual content (i.e., visual, auditory, tactile, olfactory, and gustatory) influence review usefulness; 2) to examine the multimodal sensory elements in images and uncover their impact on the usefulness of reviews; and 3) to identify the effects of discrepancies between sensory elements in text and images within multimodal online review on review usefulness. To achieve these objectives, advanced image analysis models, text mining methods and Multimodal Large Language Models (Multimodal LLMs) are employed to analyze the sensory elements in multimodal online reviews, utilizing data from TripAdvisor on hotels in Hong Kong to explore the specific sensory impacts. This research provides significant contributions. On one hand, it explores how text and images influence the usefulness of reviews from a multisensory perspective, providing a new theoretical framework for understanding multimodal online reviews. On the other hand, this study offers a novel strategy for hotels, assisting hotel managers in more effectively attracting customers through multisensory marketing.

2. Literature review

2.1. Multisensory elements in multimodal online reviews

As the influence of multimodal online reviews on consumer decision-making grows, the importance of multisensory elements is increasingly recognized by scholars and practitioners alike (Krishna & Morrin, 2008; Mattila & Wirtz, 2001; Yorkston & Menon, 2004). Multisensory elements are central to the construction of perceptual experiences (Agapito et al., 2013; 2014; Goldstein, 2010), as individuals acquire them through interactions with their environment via sensory channels such as visual, olfactory, auditory, gustatory, and tactile, as illustrated in Table S1 in the Appendix. Recent studies (Fürst et al., 2021; Krishna et al., 2010; Mehraliyev et al., 2020) have demonstrated that these sensory dimensions play a role in enhancing consumer information processing, emotional resonance, and attitude formation. Nonetheless, existing research largely focuses on sensory cues in text, with relatively limited exploration of the sensory dimensions conveyed through images. Images provide immediate visual information that enhances contextual perception of the product, compensating for the limitations of text in conveying concrete representations (Li et al., 2022). Therefore, extracting and integratively analyzing multisensory elements in both text and images not only deepens the understanding of multimodal content but also serves as a valuable tool for assessing the potential impact of multimodal reviews on consumer behavior.

Visual elements are the most direct and common sensory stimuli in multimodal online reviews (An et al., 2020; Biswas, 2019; Hamacher & Buchkremer, 2022; Kim & Chung, 2020). According to multimedia learning theory (Mayer, 1997), combinations of visual elements such as color, texture, aesthetic quality can effectively enhance learning and memory. This high-quality design of visual combinations can enhance the attractiveness and comprehensibility of information, thereby improving consumers' experience and satisfaction (Narangajavana Kaosiri et al., 2019). Li, Kwok, et al. (2023) found that the visual quality of images, including color saturation, contrast, and layout, significantly impacts customers' attention and emotions. For example, high contrast and bright colors can enhance visual usefulness, while an organized layout helps in the clear communication of information (Lim & Kim, 2020; Mayer, 1997). Additionally, visual storytelling, which narrates a story through a sequence of images, has been proven to be an effective method to increase customer engagement (Lim & Childs, 2020). While visual elements serve as the central component, the sensory experiences conveyed through textual content also hold significant importance and warrant attention (Mehraliyev et al., 2020). In line with conceptual metaphor theory (Gibbs, 2011), language-based metaphors can trigger sensory and emotional responses, even without direct sensory input. This means that textual content, through carefully constructed sensory language, can evoke sensory imagery in the audience's minds, creating an experience that is both intuitive and immersive. Therefore, given that images and text serve as joint carriers of visual elements, collaboratively exploring their role in sensory information transmission is a key means of enhancing the perceived value of the consumer.

Furthermore, sensory marketing theory (Krishna, 2012; Schmitt, 1999) highlights that the importance of auditory elements in influencing consumer perception and behavior is comparable to that of visual elements. In multimodal online reviews, auditory elements have gradually become a critical factor influencing consumer decision-making. These

elements are typically conveyed through text and image content, creating a more immersive and persuasive consumer perception. Auditory perception in textual content usually refers to the sound experience conveyed by linguistic descriptions or onomatopoeia (Russell, 2002). Such descriptions can guide potential consumers to form corresponding auditory impressions in their minds, indirectly experiencing the auditory characteristics of the product or service. Several studies have confirmed the significant value of auditory elements in textual content (Mattila & Wirtz, 2001; Mehraliyev et al., 2020; Russell, 2002; Spangenberg et al., 2005; Yorkston & Menon, 2004). For instance, the study by Mehraliyev et al. (2020) utilized text mining and sentiment analysis methods to reveal the critical role of auditory elements in users' positive and negative experiences, particularly in the impact of social media reviews on consumer ratings. Auditory perception in image reviews refers to auditory cues indirectly derived by consumers through visual content (Li et al., 2019; Sun et al., 2025). For instance, the study by Sun et al. (2025) delved into the impact of auditory elements in marketing and review images on consumers' sensory experiences, further uncovering their critical role in boosting hotel sales. It can thus be inferred that auditory perception is no longer confined to textual content; auditory elements in images have also become an integral part of influencing consumers' perceived value, although research in this area is still in its early stages.

Tactile, as the most enduring sensory modality, plays a crucial role in consumer cognition and decision-making, encompassing both explicit and implicit stimuli (Krishna, 2012; Schmitt, 1999). Particularly, within the tourism and hospitality industries, tactile influence extends beyond physical contact's explicit sensory perception, also generating psychological effects through implicit stimuli that influence consumer behavior patterns. From the perspective of explicit touch (i.e., physical contact), the product's physical characteristics, such as texture, temperature, and hardness, directly influence consumers' perceptions and judgments (Krishna et al., 2010; Krishna & Morrin, 2008). For example, the texture of mattresses, pillows, and bed linens in hotel rooms is often considered a key determinant of guest comfort. Several studies (Krishna, 2012; Krishna et al., 2010; Krishna & Morrin, 2008; Peck & Childers, 2008) have confirmed that touch effects can significantly enhance consumers' sensory experiences. Implicit touch refers to tactile associations triggered by sensory stimuli or psychological cues that influence consumers' cognition and decision-making without actual physical contact (Mehraliyev et al., 2020). For instance, innovative dish presentations in restaurants (such as hand-held rice balls or small snacks) can enhance customers' dining experience and satisfaction through intangible tactile associations (Manthiou, 2020).

In multimodal online reviews, tactile perception influences consumers' cognition and decision-making processes through forms such as images and text, with implicit tactile experiences being of equal importance as explicit tactile effects (Donato & Raimondo, 2021). For example, hotel services and design convey tactile information to potential consumers through images, advertisements, and other media, while real consumers also express their tactile experiences through text, images, and other forms. This approach offers the advantages of being low-cost and highly efficient. Implicit tactile perception has received limited attention in the tourism and consumer behavior literature, although such attention is growing (Mehraliyev et al., 2020). However, the focus has primarily been on tactile perception in a single modality (e.g., text). This study aims to explore how multimodal online reviews (i.e., text and images) convey implicit tactile perception and analyze their role in consumers' perceived usefulness of reviews.

However, sensory marketing theory highlights that consumers' perception of review value is a multidimensional and highly integrated process, wherein multiple senses—including visual, auditory, tactile, gustatory, and olfactory—interact to shape perceptual experiences (Krishna, 2012; Schmitt, 1999). While gustatory and olfactory perceptions exert a relatively limited direct influence on consumer perception compared to other sensory elements, studies have demonstrated that, in multimodal online review environments, textual and visual content can indirectly stimulate gustatory and olfactory perceptions (Biswas et al., 2021). This indirect stimulation can significantly enhance consumer satisfaction (Anguera-Torrell et al., 2021; Elder & Krishna, 2010). For instance, Elder and Krishna (2010) demonstrated that food advertisements regulate consumers' gustatory perception by influencing sensory cognition. Mehraliyev et al. (2020) employed text mining techniques and a sensory vocabulary list (such as categorical terms) to assess the impact of positive and negative gustatory and olfactory experiences on customer ratings. Nevertheless, research on gustatory and olfactory perception in multimodal online reviews remains insufficient. Text and images, as two independent media, stimulate consumers' gustatory and olfactory associations in different ways. For example, the colors and food elements in images can indirectly trigger consumers' olfactory and gustatory perceptions through visual stimuli. Therefore, clarifying the distinct roles of gustatory and olfactory perception in text and images will help enhance the practical value of multimodal online reviews and significantly strengthen their impact on the consumer decision-making process.

2.2. Text–image multisensory discrepancy

Text-image multisensory discrepancy refers to the inconsistency in sensory experience conveyed by text and image, which are two distinct modalities of information. For example, in online user reviews, the textual description might convey one sensory experience, while the accompanying image may convey a different sensory message. This discrepancy reflects a cross-modal integration barrier in consumers' perception and cognition processes, which may significantly impact the effectiveness and acceptance of information transmission (Lindgaard et al., 2006; Sun et al., 2024).

Earlier studies have conducted preliminary exploration in this field. Krishna et al. (2010) examined how the presence of olfactory elements in tactile perception contributes to sensory experience and aesthetic pleasure. They found that the semantic associations between touch and smell, along with the consistency of these associations, can significantly enhance product evaluations. Helmefalk and Hultén (2017) investigated the effects of incorporating multisensory congruent cues, such as auditory and olfactory elements, in relation to store design, product categories, and store image. They found that in a visually dominant store, adding more congruent multisensory elements (e.g., auditory and olfactory) can create a more attractive retail environment for shoppers. Fürst et al. (2021) explored the relationship between sensory attributes and product functionality. They discovered that sensory attribute-function congruence has a non-linear, positive impact on product evaluation, with either diminishing or increasing effects. Only when selected sensory attributes match the product's primary function can high sensory congruence be achieved. Most of these studies emphasize the importance of single-sensory interactions (e.g., sensory attributes and product functionality) while overlooking the effects of cross-modal.

Multisensory cross-modal interaction involves the integration and mutual influence of sensory information from different modalities, such as visual, auditory, olfactory, tactile, and gustatory. Previous research on sensory congruence has examined various cross-modal interactions across these sensory dimensions. Examples include studies exploring the interplay between olfactory and auditory stimuli (Mattila & Wirtz, 2001), auditory and visual (Russell, 2002), auditory and olfactory (Spangenberg et al., 2005), auditory and gustatory (Yorkston & Menon, 2004), tactile and gustatory (Krishna & Morrin, 2008), visual and gustatory (Hoegg & Alba, 2007), as well as multisensory cognition and gustatory perception (Elder & Krishna, 2010). These findings emphasize the interactive dynamics of sensory information. For instance, Mattila and Wirtz (2001) found that in retail environments, the combination of scent and music promoted approach behaviors, particularly when both sensory cues exhibited congruent arousal levels.

Although these studies (e.g., Elder & Krishna, 2010; Krishna et al., 2010; Mattila & Wirtz, 2001; Mehraliyev et al., 2020) have measured the effects of various senses (e.g., visual, auditory, olfactory, tactile, and gustatory) on user ratings and experiences, they primarily focus on data consistency within the same modality, overlooking the impact of text-image multisensory discrepancy. This discrepancy arises because sensory perception is a fundamental basis for human rational choice. When faced with information from different modalities (e.g., text and image), people often exhibit systematic cognitive biases (Hoegg & Alba, 2007; Sun et al., 2024). These biases stem from the unique characteristics of each sensory channel, which individually affect consumers' understanding of reviews and their immersive experience (Li et al., 2019, 2022; Mehraliyev et al., 2020). However, there is a lack of research on multimodal online reviews that captures sensory experiences across different dimensions and examines the impact of sensory discrepancies between various modalities.

The necessity of studying text-image multisensory discrepancy lies in its ability to help us understand the complex impact of multimodal information on consumer cognition and perception. In the process of information processing, humans often rely on the integration of visual and textual information to form a comprehensive cognitive experience (Ahtola, 1985; Dewey, 1934). However, when information inconsistency arises between different modalities, consumers may experience cognitive dissonance, affecting both the perceived credibility of the information and the enjoyment of the experience (Festinger, 1959). Inconsistency in multimodal information may cause consumers to develop biased perceptions of a product, subsequently affecting their decision-making process. Studies by Li et al. (2019, 2022) also demonstrate that cross-modal information discrepancies can influence users' immersive experiences and overall trust in reviews. Therefore, studying text-image multisensory discrepancies not only contributes to a deeper understanding of the complexities of information transmission but also provides a theoretical foundation for the fields of digital marketing and consumer psychology.

2.3. Significance of multisensory elements

Sensory marketing theory emphasizes that consumer decision-making is influenced not only by rational factors but also shaped significantly by sensory experiences (Krishna, 2012; Lee et al., 2019; Sun et al., 2025). In traditional marketing research, consumers are primarily regarded as rational information processors (Li et al., 2022), whereas sensory marketing emphasizes the use of multiple sensory stimuli, including visual, auditory, olfactory, gustatory, and tactile cues, to evoke consumer responses and purchase motivation. For consumers, multisensory elements in text and images not only provide a rich presentation of information but also enhance the perceived value of the product by activating different sensory channels (Sun et al., 2025). This multisensory integrated experience can improve the efficiency of information processing, thereby influencing consumer decision-making (Mehraliyev et al., 2020; Sun et al., 2025).

2.3.1. Text multisensory elements and consumers' perceived value

The importance of review usefulness lies in its significant impact on potential consumers' perceptions, preferences, and purchasing decisions, making it a key factor in influencing consumers satisfaction (Fang et al., 2019; Li et al., 2022, 2023b). Multisensory elements embedded in multimodal online reviews, particularly textual content, play a critical role in enhancing the perceived usefulness of reviews and influencing potential consumers' purchase intentions (Andersson, 2014; Lee et al., 2010; Lyu & Huang, 2024).

According to the theory of sensory marketing (Krishna, 2012; Schmitt, 1999), textual descriptions can effectively engage consumers' multisensory experiences. Although textual content is inherently descriptive, it can evoke associations with other senses (visual, auditory, tactile, gustatory, and olfactory) through detailed descriptions. This sensory stimulation is primarily achieved through: detailed descriptions of colors, shapes, and spatial characteristics to evoke visual experiences (Mehraliyev et al., 2020); depictions of onomatopoeia or sound characteristics to arouse auditory associations (Russell, 2002); descriptions of textures or tactile sensations to stimulate touch perception (Krishna & Morrin, 2008); and portrayals of food or aromas to enhance olfactory and gustatory imagination (Mattila & Wirtz, 2001; Yorkston & Menon, 2004). This language-driven multisensory experience enhances the influence of textual content on consumers' cognitive and emotional engagement. For instance, a consumer wrote in their review: *"The hotel's environment design is very exquisite, the lighting is warm, and a faint fragrance fills the hotel. The restaurant's food is both delicious and visually appealing, leaving me very satisfied and relaxed by the end"* (https://www.tripadvisor.com.au/). This review authentically reflects the consumer's experience, represented through multisensory elements such as visual, olfactory, gustatory, and tactile. These descriptions influence consumers' perception of review value through multiple sensory channels (Mehraliyev et al., 2020). The information conveyed through each sensory channel can evoke consumers' associations and experiences with the product or service, thereby enhancing the perceived value of the review on multiple levels. For example, Archak et al. (2011) noted that information-dense reviews can increase product visibility and attractiveness, thereby directly influencing purchase intentions. Cheung and Thadani (2012) further demonstrated that sensory-rich text can enhance the vividness of product descriptions and potential consumer experiences, which is crucial for making final purchasing decisions. Therefore, the textual content of online reviews, by transmitting multisensory cues, can influence consumers' perception of the value of the review to a certain extent. Therefore, the first research question of this study is.

Q1. *How do text multisensory elements influence consumers' perceived usefulness of reviews?*

2.3.2. Image multisensory elements and consumers' perceived value

Visual content serves as a crucial medium for conveying complex information and plays a key role in communication processes (Li et al., 2023a, 2023b, 2024a). According to the media richness theory (Mayer, 1997), visual content, compared to text, leverages its advantages in vividness and information density to communicate complex information more effectively and offer consumers a more immersive experience (Li, Liu, et al., 2024; Zhu et al., 2025). This high-information-density characteristic enables visual content to present rich details within a limited time, thereby enhancing consumers' comprehension and perception. Building on the Mental Imagery Theory (Kosslyn, 1996), visual content not only conveys information directly but also activates consumers' mental imagery, allowing them to visualize and simulate experiences in their minds. This mental simulation enhances the depth of information processing and strengthens the cognitive impact of visual content.

However, the impact of visual content is not confined to mere visual transmission (Sun et al., 2025). According to sensory marketing theory (Krishna, 2012; Schmitt, 1999), visual content not only captures consumers' attention through its rich visual elements but also triggers a multisensory response, rapidly stimulating a comprehensive experiential and cognitive reaction from consumers. Mental Imagery Theory further supports this by suggesting that visual content can evoke vivid mental images that extend beyond the visual domain, engaging other sensory modalities such as olfactory, auditory, gustatory, and tactile sensations (Kosslyn, 1996; Sun et al., 2025). For instance, Sun et al. (2025) demonstrated that multisensory elements (olfactory, auditory, and tactile) embedded in review images and marketing images can significantly evoke consumers' perceptual experiences, thereby influencing their decision-making processes. Therefore, these multisensory elements enable visual content to transcend static dimensions, becoming an integral part of consumers' perceptions and experiences. They not only enhance the appeal of visual content but also elevate consumers'

overall perception of hotels and their services.

As consumer demands continue to evolve, the role of multisensory elements in images has become increasingly prominent in shaping consumer experiences. These elements are not only direct reflections of actual consumer experiences but also crucial factors in shaping potential consumers' perceptions of brand value (Sun et al., 2025). Research by Gentile et al. (2007) and Schmitt (1999) suggests that multisensory elements can simulate real sensory experiences, thereby enhancing potential consumers' interest and sense of identification. Moreover, multisensory elements can reflect genuine consumer sentiments in reviews, further influencing other potential consumers' trust in the brand and their decision-making tendencies (Li et al., 2011).

The realization of multisensory stimulation primarily depends on the richness of visual content and its interactive effects (Sun et al., 2025). Through visual presentation, images can not only directly stimulate consumers' visual perception but also indirectly evoke associations with other senses. For instance, showcasing a quiet lobby scene can evoke consumers' imagination of auditory comfort; presenting the softness of room fabrics can stimulate tactile experiences; and the delicate presentation of food can resonate with gustatory and olfactory senses. This multisensory interactive communication enables visual content to transcend mere image representation, becoming a critical bridge between consumer perception and brand interaction. Therefore, the multisensory elements embedded in visual content are key drivers in facilitating consumers' simulated experiences and shaping their perceived value of reviews. Accordingly, the second research question of this study is.

Q2. *How do image multisensory elements influence consumers' perceived usefulness of reviews?*

2.3.3. Text-image multisensory discrepancies and consumers' perceived value

The interaction between textual and visual information has long been a central issue in the fields of emotional communication and consumer cognition (Daft & Lengel, 1986). According to the dual-coding theory (Paivio, 1990), textual information is encoded through the verbal system, while visual information is processed through the non-verbal system. These two pathways exhibit significant discrepancies in terms of information encoding, storage, and retrieval. For example, Li et al. (2022) examined the impact of discrepancies in emotional expression between review text and images on review usefulness and review enjoyment, highlighting the critical role of text-image emotional discrepancies in shaping consumer behavior. These findings suggest that variations in the nature of media transmission can lead to significant discrepancies in consumers' information processing methods, which, in turn, become key factors influencing consumer perception and decision-making processes (Daft & Lengel, 1986; Lee & Choi, 2019; Li, 2022; Li et al., 2022).

However, according to sensory marketing theory (Krishna, 2012; Schmitt, 1999), the sensory discrepancies between text and images are also critical factors influencing the consumer perception process. Textual information typically relies on cognitive processing, characterized by logical presentation and linear thinking, enabling detailed explanations and specific information (Mayer, 2005). In contrast, visual information primarily engages affective processing, quickly eliciting sensory experiences through intuitive visual cues (Li et al., 2022). When significant sensory discrepancies exist between text and images, they can increase consumers' cognitive load (Sweller, 1988), leading to phenomena such as inconsistent sensory experiences. Such inconsistency can significantly hinder consumers' perception of the value of reviews. Research in sensory marketing suggests that the congruence of multisensory information enhances consumers' memory and trust in the information, while discrepancies or conflicts may prompt reflective evaluation, increasing the difficulty of judging the information's credibility (Elder & Krishna, 2010; Krishna & Morrin, 2008; Mattila & Wirtz, 2001; Yorkston & Menon, 2004). Therefore, the multisensory discrepancies between text and images affect consumers' intuitive perception of reviews, which in turn influences the actual usefulness of the reviews in their decision-making process. Accordingly, the third research question of this study is.

Q3. *How does text-image multisensory discrepancy influence consumers' perceived usefulness of reviews?*

Our research framework is summarized in Fig. 1.

3. Research methodology

3.1. Data collection

According to the report by the International Hotel and Restaurant Association (IHandRA, 2023), the hotel industry plays a crucial role in the global economy. This sector encompasses not only hotels but also includes catering services, conference and event organization, and other components. Data shows that the hotel industry contributes $3.4 trillion to the global economy annually. Focusing on Hong Kong, China, the hospitality's investment in 2023 is expected to increase by 24 %, reaching about $1 billion. This significant growth indicates a recovery to the levels of 2018 by 2024 (Jones Lang LaSalle Incorporated, 2023). Given this, we have chosen to focus our research on the hotel industry, with a particular emphasis on Hong Kong, China. We collected data from the top 5 five-star hotels ranked by the number of reviews on TripAdvisor, namely Four Seasons Hotel Hong Kong (A1), Hotel ICON (A2), Mandarin Oriental Hong Kong (A3), The Ritz-Carlton Hong Kong (A4), and W Hong Kong (A5). TripAdvisor (https://www.tripadvisor.com.au/) is one of the world's most popular online review platforms, attracting 340 million unique visitors monthly, with over 78 million registered members and more than 2 trillion reviews (TripAdvisor, 2023). Hotel marketers can describe their services and features on this platform, while consumers can write online reviews about their lodging experiences, as shown in Fig. 2.

Our data comprises three main dimensions: 1) hotel dimension, which includes the hotel's name, hotel ID, and related attributes; 2) review dimension, encompassing review ID, reviewer's name, ID, review date, review experience rate, review helpfulness score, review text, and review title; 3) review image dimension, comprising the corresponding review ID, image ID, and image file. We grouped all the online reviews by reviewer ID, treating each reviewer as an individual unit, and ultimately identified 7333 unique reviewers (Chatterjee, 2020; Liang et al., 2019). Along with their corresponding 23,198 images, we constructed a daily unbalanced panel dataset covering the reviews of each hotel up to October 22, 2023, based on this data.

3.2. Variable measurement in research framework

The variable measurement in this study is divided into three main categories: dependent variables, independent variables, and control variables. The study defines these variables by extracting characteristic attributes from both text and image. The detailed framework for variable measurement is presented in Fig. 3.

3.2.1. Dependent variables

Our dependent variable was the perceived review usefulness. Review usefulness was measured by a review's number of "helpful" votes, and this aligns with the existing literature (Li et al., 2022; Mohammed et al., 2019; Zhang et al., 2011; Zhu et al., 2025).

3.2.2. Explanatory variables

The independent variables of this study are sensory elements, which include two aspects: textual content and image content. The textual sensory elements are *TEX_Visual*, *TEX_Auditory*, *TEX_Tactile*, *TEX_Gustatory*, and *TEX_Olfactory*. They are defined as the proportion of words

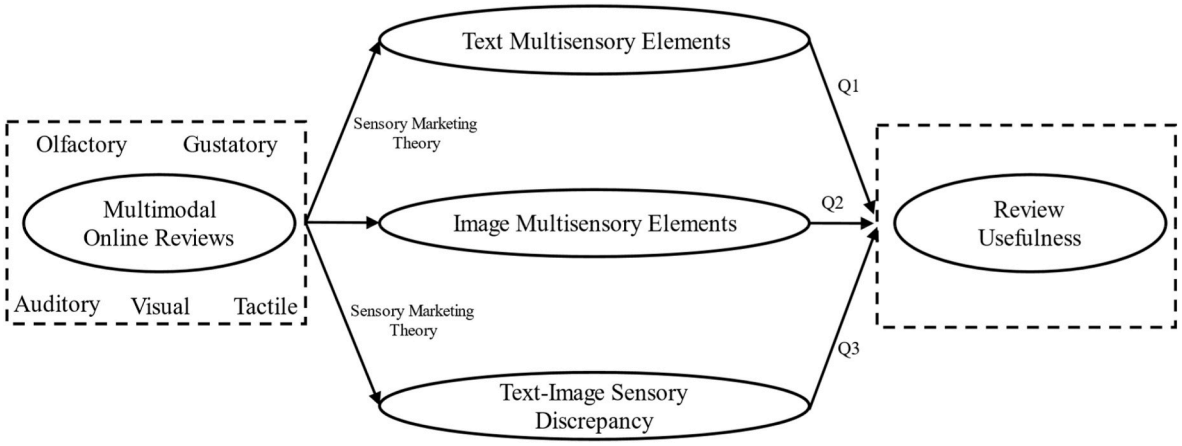


Fig. 1. Research framework.

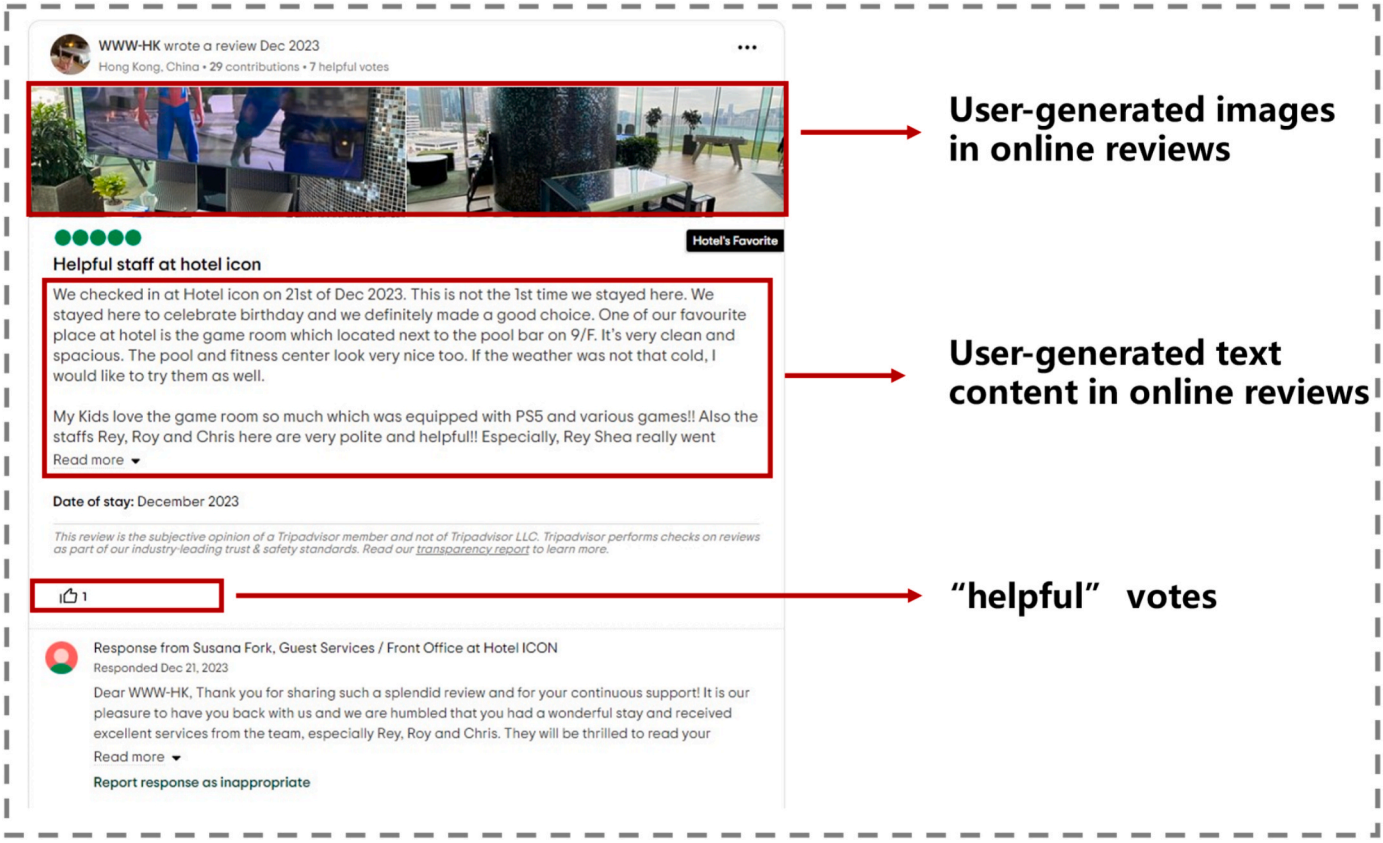


Fig. 2. Example of the user-generated content in multimodal online reviews.

related to visual, auditory, tactile, gustatory, and olfactory senses in the total words of online text reviews. The image multisensory elements include *IMA_Visual*, *IMA_Auditory*, *IMA_Tactile*, *IMA_Gustatory*, and *IMA_Olfactory*. They are defined as the proportion of sensory dimensions—visual, auditory, tactile, gustatory, and olfactory—within the total information converted from the images in each review.

For textual multisensory elements, this study combines an *English Sensory Dictionary* and *Word2Vec* for measurement. Specifically, drawing on the sensory-emotion word set constructed by Yan et al. (2023) and the approach to constructing textual indices by Hu et al. (2021), this research identifies a set of sensory feature words from text reviews. As expressions of the same concept or object often involve multiple semantically similar words, this research uses *Word2Vec* to expand the seed word set with similar words. *Word2Vec* is a word embedding technology for natural language processing developed by Google (https://www.google.com/), employing a shallow neural network model that represents words as vectors in a continuous vector space. Its core idea is to map vocabulary in a language into a high-dimensional vector space, where semantically similar words are proximate. Trained on large-scale text corpora, the *Word2Vec* model can learn the semantic relationships between words, ensuring that words with similar contexts in vector space are close to each other. Based on the seed word set, we identified approximately 500 semantically similar words. Since *Word2Vec* is trained on massive volumes of review texts, the recommended similar words are more suited to the context of review texts, effectively avoiding the subjectivity of manually defined word lists and

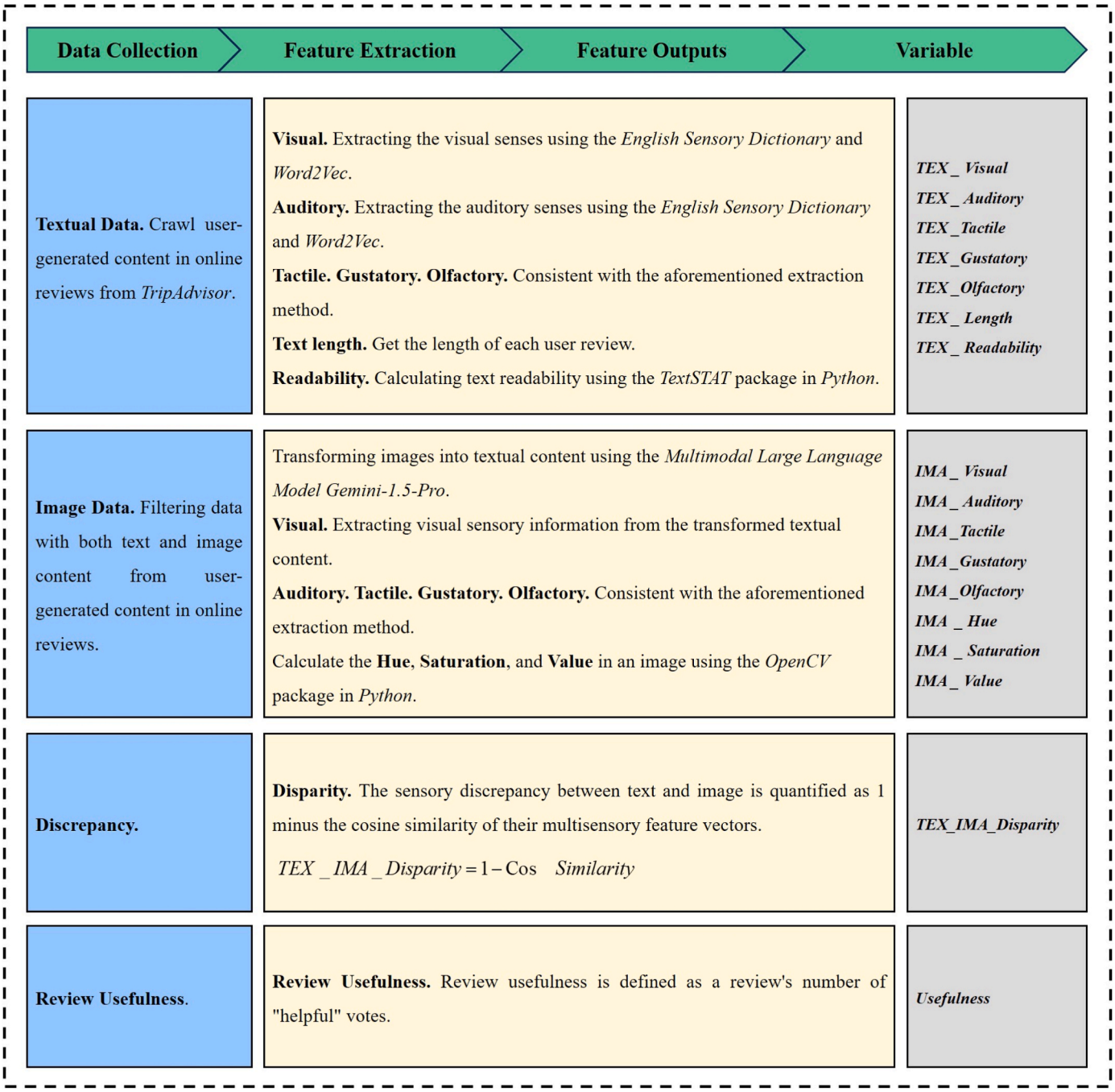


Fig. 3. Variable measurement in research framework.

the weak relevance of generic synonym tools. Finally, we invited three doctoral students with related research experience to verify the list. After manual filtering and deduplication, we obtained 471 visual dimension sensory words, 482 auditory, 437 tactile, 502 olfactory, and 498 gustatory words. We then calculated the similarity of each review to the words in the sensory dictionary, as shown in Equation 1.

$$Similarity(i,j,k) = \frac{\overrightarrow{v_{ij}} \cdot \overrightarrow{w_k}}{\left\|\overrightarrow{v_{ij}}\right\| \cdot \left\|\overrightarrow{w_k}\right\|} \quad (1)$$

Where $\overrightarrow{v_{ij}}$ represents the j-th word in the i-th review, and $\overrightarrow{w_k}$ represents the k-th word in the sensory dictionary. The dot product "·" indicates the inner product of two vectors, and $\left\|\overrightarrow{v_{ij}}\right\|$, $\left\|\overrightarrow{w_k}\right\|$ represent the magnitudes of the vectors, respectively.

According to previous research (Li et al., 2021; Olson et al., 2021; Packard & Berger, 2021), when the similarity exceeds 0.6, it can be approximately assumed that word j and the sensory dimension represented by word k in the sensory dictionary are the same. Ultimately, the proportion of occurrences of a particular sensory dimension in a review is calculated as a ratio of the total number of words to derive the textual sensory element variable.

Recently, Multimodal LLMs have demonstrated exceptional performance in integrating vision and language for multimodal tasks. Compared to traditional models, Multimodal LLMs combine robust language comprehension with advanced visual processing capabilities. They have achieved significant progress in areas such as image classification, object detection, and generative tasks (e.g., image generation and image-text translation) while excelling in handling complex cross-

modal description tasks. These models are capable of generating highly detailed and natural descriptions, converting critical information from images into semantically accurate and linguistically fluent text. They can not only capture obvious visual information, such as object categories, quantities, and positions, but also describe deeper semantic content such as scene backgrounds, emotional atmospheres, and dynamic actions. For instance, when analyzing a portrait, the model can not only identify "*this is a woman*" but also provide a detailed description such as "*she is wearing a blue dress, holding an open book, smiling while gazing into the distance, with a sunny park as the background.*"

Therefore, in the analysis of multisensory elements in images, this study employed the image-to-text conversion method based on the Gemini-1.5-Pro model to achieve semantic understanding and text generation of image data. Gemini-1.5-Pro is the latest Multimodal LLMs developed by Google (https://ai.google.dev/gemini-api/docs). The data analysis was conducted from November 2, 2024, to November 11, 2024, during which we utilized the Gemini-1.5-Pro model's API to process and analyze image data, as illustrated in Fig. 4. The model first performs fine-grained feature extraction of image information through its enhanced visual module and then converts these visual features into semantically rich and coherent textual descriptions via its language generation module. The technical details of the processing steps are provided in Appendix Table S3. Using this method, we generated textual descriptions of the image information. As shown in Fig. 5, we demonstrate the comprehension results for three different types of image content.

Then, through Equation (1), we calculate the similarity of each word to every word in the sensory dictionary. When the similarity exceeds 0.6, it is considered that the word matches the sensory characteristic represented by the corresponding sensory word (Li et al., 2021; Olson et al., 2021; Packard & Berger, 2021). Next, we calculate the proportion of each sensory dimension in the textual content converted from the images attached to each review, to derive the image sensory element variable. Finally, we calculated the average percentage of each textual sensory dimension and the average percentage of each visual sensory dimension across all the reviews posted by each reviewer (Chatterjee, 2020; Liang et al., 2019).

We further measured the sensory discrepancy between text and images using cosine similarity. First, we used Equation (2) to evaluate the directional similarity between text and image variables.

$$CosSimilarity = \frac{A \cdot B}{\|A\| \cdot \|B\|} \tag{2}$$

Where A and B represent the sensory feature vectors of text and images, respectively, and $\|A\|$ and $\|B\|$ are the Euclidean norms of the corresponding vectors.

Next, based on cosine similarity, the sensory discrepancy between text and images is defined by Equation (3).

$$TEX_IMA_Disparity = 1 - CosSimilarity \tag{3}$$

3.2.3. Control variables

Consumers can browse past reviews of hotels on websites, which include the upload date of the review, the length of the text review, and the richness of color in image reviews. These pieces of information can consciously or unconsciously influence review usefulness and customer experience. Therefore, we controlled key elements influencing review usefulness in the model that could confound the estimation results. First, we define *TEX_Length* as the length of the text review. Generally, the longer the review, the richer the product information it contains, and the higher its usefulness. Second, *TEX_Readability* is defined as the readability of the text review. Previous research (Graesser et al., 2004; Klare, 1974; Sung et al., 2016) has shown that the higher the readability of a review, and the easier it is for customers to understand. Third, *IMA_Hue* is defined as the average hue value in the *HSV* color space of each image attached to a review, *IMA_Saturation* as the average saturation value in the *HSV* color space, and *IMA_Value* as the average brightness value in the *HSV* color space. Earlier studies (Barnes, 2022; Filieri et al., 2021; Hou et al., 2023; Siamionava et al., 2018) have shown that the color of images is extremely important in forming customers' first impressions. Lastly, we controlled both time fixed effects, accounting for variations due to upload dates, and hotel fixed effects. Regarding the variable calculation, *TEX_Length* is calculated using Python. *TEX_Readability* is estimated using TextSTAT, a Python-based text statistics computation library. *IMA_Hue*, *IMA_Saturation*, *IMA_Value* are calculated using the OpenCV package. OpenCV provides a wealth of image processing capabilities, including handling color spaces, allowing us to efficiently gather information about the Hue, Saturation, and Value of images.

The descriptive analysis of the above variables is shown in Table 1. Correlation analysis was performed among all independent and control

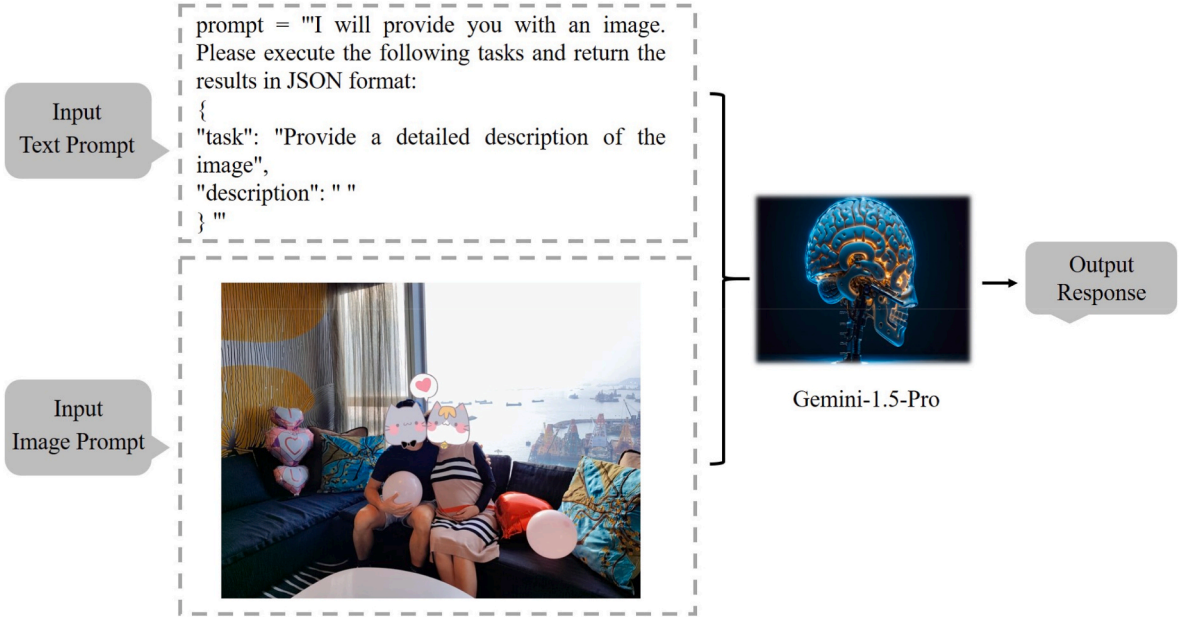


Fig. 4. Process of using large language models to analyze image content.



Fig. 5. Illustrative examples of image content analysis.

variables as displayed in Fig. 6, revealing no collinearity issues given these small correlation coefficients.

3.3. Model specification

To examine the impact of multisensory elements on *Review Usefulness*, Poisson regression, a type of generalized linear model (GLM), was employed since *Review Usefulness* is a count variable (Li et al., 2022; Mohammed et al., 2019; Zhu et al., 2025).

The impact of text and image multisensory elements on review usefulness was first tested separately, with the specific econometric model being **Model 1** and **Model 2**, respectively.

$$Usefulness = \alpha_1 + \beta_1 Text_Sensory + \pi_1 Control \quad (1)$$

$$Usefulness = \alpha_2 + \beta_2 Image_Sensory + \pi_2 Control \quad (2)$$

Where β_1, β_2, and α_1, α_2 represent the parameter estimate and constant terms, respectively. "*Text_Sensory*" refers to the multisensory elements in the text, including visual, auditory, tactile, olfactory, and gustatory aspects. "*Image_Sensory*" refers to the multisensory elements in the images, encompassing visual, auditory, tactile, olfactory, and gustatory aspects. "*Control*" refers to the control variables, such as text length, text readability, image color, date, and others.

Model 3. was proposed to further measure the impact of review text and image sensory elements on review usefulness simultaneously. The specific econometric model is as follows:

$$Usefulness = \alpha_3 + \beta_{31} Text_Sensory + \beta_{32} Image_Sensory + \pi_3 Control \quad (3)$$

Where β_{31}, β_{32}, and α_3 represent the parameter estimates and the constant term, respectively.

Model 4. was proposed to further measure the impact of the

Table 1
Descriptive analysis.

Variable	Obs	Mean	Std. dev.	Min	Max
Usefulness	7333	3.379	6.561	0.000	75.00
TEX_ Visual	7333	0.211	0.055	0.000	0.515
TEX_ Auditory	7333	0.019	0.018	0.000	0.139
TEX_ Tactile	7333	0.085	0.039	0.000	0.371
TEX_ Gustatory	7333	0.044	0.036	0.000	0.429
TEX_ Olfactory	7333	0.019	0.020	0.000	0.371
IMA_ Visual	7333	0.312	0.052	0.000	0.583
IMA_ Auditory	7333	0.034	0.019	0.000	0.111
IMA_ Tactile	7333	0.112	0.039	0.000	0.300
IMA_ Gustatory	7333	0.039	0.031	0.000	0.215
IMA_ Olfactory	7333	0.036	0.023	0.000	0.200
TEX_IMA_Disparity	7333	0.037	0.078	0.000	1.000
TEX_ Length	7333	716.6	666.73	130.0	26736
TEX_ Readability	7333	8.336	3.0330	2.000	69.000
IMA_H	7333	55.85	24.403	5.914	150.785
IMA_S	7333	83.39	26.542	3.199	240.029
IMA_V	7333	125.157	28.202	14.936	229.943

discrepancy between textual and image sensory elements on review usefulness. The specific econometric model is as follows:

$$Usefulness = \alpha_4 + \beta_{41} Text_Sensory + \beta_{42} Image_Sensory + \beta_{43} TEX_IMA_Disparity + \pi_4 Control \quad (4)$$

Where, β_{41}, β_{42}, β_{43} and α_4, represent the parameter estimates and the constant term, respectively. "*TEX_IMA_Disparity*" refers to the discrepancy between textual and image sensory elements.

4. Analysis of results

4.1. Sensory elements of multimodal online reviews on review usefulness

To evaluate the influence of textual sensory elements on review usefulness, we conducted Poisson regression, as shown Table 2 (**Model 1**). The results show that *TEX_Auditory* ($\beta = 1.803$, $p < 0.01$), *TEX_Tactile* ($\beta = 2.787$, $p < 0.01$), and *TEX_Gustatory* ($\beta = 3.737$, $p < 0.01$) positively influenced usefulness.

The positive impact of auditory descriptions can be attributed to their ability to evoke mental imagery of sounds, helping consumers imagine specific scenarios such as the ambiance of a cafe or the hustle of a marketplace. This immersive simulation reduces uncertainty by making abstract textual content more relatable and vivid. Similarly, tactile descriptions enhance usefulness by providing concrete details about textures or sensations, allowing consumers to mentally assess the physical attributes of a product, which is particularly important when direct interaction is not possible. Gustatory descriptions are the most impactful because they vividly translate flavors and textures into relatable sensory information, fulfilling consumers' need for specificity when evaluating food and beverage products. These effects align with sensory marketing theory (Krishna, 2012; Schmitt, 1999), which posits that sensory descriptions enable consumers to mentally simulate experiences, bridging the gap between abstract text and tangible interactions. Furthermore, Conceptual Metaphor Theory (Gibbs, 2011) explains how sensory language fosters intuitive and engaging connections, enhancing the overall persuasiveness of the review.

Our regression analysis revealed that visual content proportion showed a significantly negative impact on review usefulness ($\beta = -1.629$, $p < 0.01$). This can be explained by three fundamental mechanisms: First, from the perspective of cognitive resource competition, the human information processing system has inherent attentional bottlenecks (Kahneman, 1973). When reviews frequently contain visual descriptions such as "crystal chandeliers" or "marble floors," such decorative information continuously consumes readers' limited working memory resources, creating cognitive interference with functional information like "mattress support" or "soundproofing effectiveness." Second, from a multisensory balance perspective, effective evaluation of

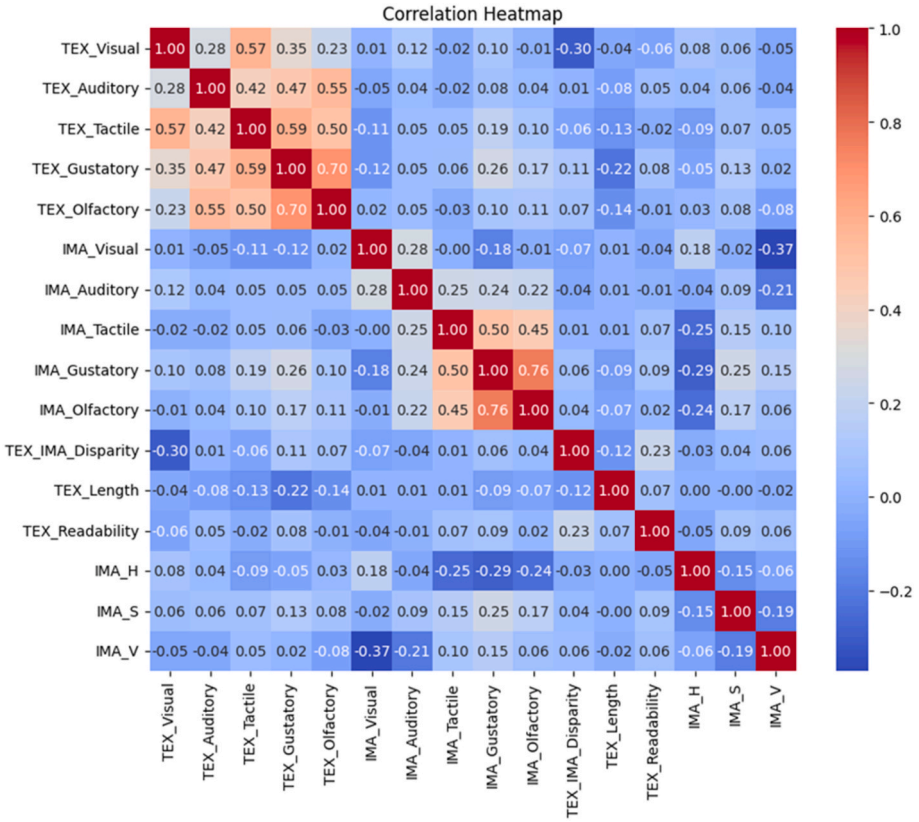


Fig. 6. Correlation analysis.

Table 2
Results of the impact of multisensory elements on review usefulness.

Variable	Usefulness			
	Model 1	Model 2	Model 3	Model 4
TEX_Visual	−1.629*** (0.137)		−1.673*** (0.140)	−2.097*** (0.150)
TEX_Auditory	1.803*** (0.380)		2.119*** (0.385)	2.248*** (0.388)
TEX_Tactile	2.787*** (0.250)		2.935*** (0.252)	2.948*** (0.254)
TEX_Gustatory	3.737*** (0.250)		3.558*** (0.256)	4.059*** (0.265)
TEX_Olfactory	−3.616*** (0.389)		−3.661*** (0.401)	−3.508*** (0.408)
IMA_Visual		−0.849*** (0.151)	−0.389** (0.151)	−0.344** (0.152)
IMA_Auditory		3.799*** (0.349)	4.065*** (0.352)	3.991*** (0.354)
IMA_Tactile		0.984*** (0.201)	0.837*** (0.205)	0.727*** (0.206)
IMA_Gustatory		−0.940*** (0.360)	−2.210*** (0.366)	−2.353*** (0.367)
IMA_Olfactory		2.909*** (0.438)	3.0213*** (0.438)	2.989*** (0.440)
TEX_IMA_Disparity				−1.275*** (0.161)
TEX_Length	0.000035*** (0.000013)	−0.000048*** (0.000014)	0.0000344** (0.0000133)	0.0000219 (0.000014)
TEX_Readability	0.0115*** (0.00237)	0.0134*** (0.00219)	0.00961*** (0.0023)	0.0128*** (0.00238)
IMA_H	0.00213*** (0.00029)	0.00187*** (0.000302)	0.00261*** (0.000304)	0.00264*** (0.00031)
IMA_S	−0.00193*** (0.000252)	−0.00237*** (0.00026)	−0.00192*** (0.000255)	−0.00189*** (0.00026)
IMA_V	−0.00224*** (0.000248)	−0.002061*** (0.000261)	−0.00184*** (0.000258)	−0.00191*** (0.00026)
Time fixed effect	Yes	Yes	Yes	Yes
Hotel fixed effect	Yes	Yes	Yes	Yes
Observations	7333	7333	7333	7333
Log-Likelihood	−17132.319	−17341.945	−17007.174	−16966.759
Pseudo R2	0.5292	0.5235	0.5327	0.5338

Note: *, **, *** indicates 10 %, 5 %, and 1 % significance level.

accommodation experiences requires coordinated presentation of information across different sensory dimensions. Research in sensory marketing (Krishna, 2012) shows that excessive emphasis on visual descriptions leads to the systematic marginalization of other sensory information such as tactile and auditory cues. For example, when reviews devote substantial space to describing decorative details, they often compress the description space for critical experience dimensions like "towel softness" or "environmental quietness." It is important to emphasize that these findings reveal structural patterns rather than absolute rules. Visual information still holds irreplaceable value in shaping first impressions, among other functions. However, when its proportion exceeds the system's equilibrium point, the aforementioned negative effects emerge. This provides important insights for optimizing the information structure of online reviews.

The results showed an even stronger negative effect for olfactory descriptions ($\beta = -3.616$, $p < 0.01$). This can be attributed to three fundamental challenges: First, scent perception is highly subjective and culturally dependent, making standardized descriptions difficult. Second, the English language lacks specific vocabulary for odors, forcing reviewers to rely on analogies (e.g., "smelled like damp towels") that may not translate consistently across readers. Third, unlike tactile experiences which can be objectively verified (e.g., texture), olfactory experiences are particularly vulnerable to individual differences in sensitivity and preference. These limitations, as noted by Mehraliyev et al. (2020), significantly reduce the reliability and practical utility of scent-related information in consumer decision-making contexts.

We examined the impact of image-based sensory cues on review usefulness, and the findings in **Model 2** reveal significant positive effects of auditory, tactile, and olfactory cues. *IMA_Auditory* ($\beta = 3.799$, $p < 0.01$) enhanced usefulness by visually representing soundscapes or noise levels, helping consumers mentally simulate auditory experiences. *IMA_Tactile* ($\beta = 0.984$, $p < 0.01$) contributed by depicting textures or surfaces, enabling consumers to infer tactile qualities like softness or roughness. *IMA_Olfactory* ($\beta = 2.909$, $p < 0.01$) evoked strong sensory associations through imagery of fragrant objects, such as flowers or spices, enriching mental representations of scent and enhancing emotional connection. These findings validate the saying, "*A picture is worth a thousand words*". Images convey sensory information more vividly than text, enabling consumers to simulate product interactions and reducing decision-making uncertainty, particularly in online contexts (Li et al., 2022).

However, not all image-related sensory elements enhanced review usefulness. Both *IMA_Visual* ($\beta = -0.849$, $p < 0.01$) and *IMA_Gustatory* ($\beta = -0.940$, $p < 0.01$) exhibited significant negative effects. While the detrimental impact of visual elements follows the same cognitive interference mechanism discussed earlier, gustatory imagery—such as food photographs—presents unique challenges. Unlike objective visual details, taste is highly subjective and difficult to convey through static images. A well-presented dish, for instance, cannot reliably communicate nuanced flavor profiles (e.g., "balanced acidity" or "umami richness"), creating a credibility gap when consumers compare depicted meals with actual experiences. This aligns with Cognitive Load Theory (Sweller, 1988), as ambiguous sensory representations force readers to reconcile mismatched expectations, increasing mental effort while eroding trust in the review's utility. Importantly, these patterns remained consistent across **Models 1–3**, confirming the stability of multisensory variable effects even when accounting for different sets of predictors.

Finally, the results in Model 4 indicate that the discrepancy between textual descriptions and image cues significantly undermined the perceived usefulness of reviews ($\beta = -1.275$, $p < 0.01$). This finding further highlights the profound impact of inconsistencies between cross-modal information (such as text and images) in multimodal online reviews (Helmefalk & Hultén, 2017; Krishna, 2010; Li, 2022; Li et al., 2022; Mattila, 2011). Specifically, when there are sensory discrepancies between textual descriptions and the information conveyed by images, consumers are likely to question the authenticity and reliability of the reviews, thereby reducing trust in them (Li et al., 2022). This indicates that consistent sensory signals play a critical role in consumers' decision-making processes, as they help provide a coherent and trustworthy informational framework. Previous studies have extensively explored the role of information consistency in influencing consumer behavior (Krishna, 2010; Li et al., 2022; Russell, 2002). For instance, Russell (2002) conducted one of the earlier studies examining the consistency between modality and plot connection, finding that while inconsistencies can significantly enhance brand memory, they may simultaneously exert a negative impact on brand attitude. Building on this foundation, Li et al. (2022) conducted an in-depth analysis of the discrepancies in emotional transmission between textual and visual cross-modal information, further revealing the significant negative impact of such inconsistencies on review usefulness and review enjoyment. This study further advances the field from a multisensory

perspective by demonstrating that sensory discrepancies between text and images have a significant negative impact on the perceived usefulness of reviews.

4.2. Robustness check

The rapid advancement of Multimodal LLMs may be one of the primary factors influencing the comprehension of image content, particularly through their ability to transform image content into text content, thereby enabling more nuanced and language-based interpretation of visual information. In our study, we employed Gemini-1.5-Pro, a Multimodal LLMs developed by Google, to extract textual content from image data. Leveraging the extracted content, we quantitatively assessed the multisensory elements embedded within the image content. To further test the robustness of our models, we employed the GLM-4V-Plus-0111 model to validate our findings. This model, developed by Zhipu AI and released on January 11, 2025, is one of the state-of-the-art Multimodal LLMs with advanced capabilities in understanding images, videos, audio, and other multimodal data (Zhipu AI, 2024).

Leveraging the GLM-4V-Plus-0111 model, we converted image data into textual representations (March 15–25, 2025) and re-quantified the multisensory elements embedded in the image content. Through empirical analysis, we estimated their impact, with detailed results presented in Table 3. The estimation results indicate that the influence of multisensory elements in both text and images on review usefulness aligns with the findings from prior analyses. Furthermore, the results reveal that the sensory discrepancy between text and images exerts a significant negative effect on review usefulness.

5. Discussion and implications

5.1. Conclusions and discussion

Given the importance of multisensory elements in multimodal online reviews, an in-depth analysis of these elements is crucial for enhancing the perceived value for potential consumers (Luo et al., 2021; Wang et al., 2018). These elements not only reflect past consumers' experiences but also deepen future/potential consumers' understanding of services by evoking immersive sensory simulations. Therefore, clarifying the specific role of multisensory elements in multimodal online reviews on consumers' perception of review usefulness holds significant theoretical implications for enhancing consumer experience (Lopez & Garza, 2022; Mathur et al., 2022). The sensory elements embedded in text and images influence consumers' simulated experiences through multisensory channels, further shaping their subjective perceptions and evaluations of review content (Li et al., 2019). These sensory stimuli not only affect consumers' understanding of review content but also indirectly influence their evaluations of review usefulness by eliciting emotional resonance. Consequently, investigating how multisensory elements in multimodal online review content impact consumers' perceived review usefulness has become a critical focus of current research.

Table 3
The robustness check results.

Variable	Usefulness
	Model 5
TEX_Visual	−1.986*** (0.149)
TEX_Auditory	1.999*** (0.387)
TEX_Tactile	2.838*** (0.256)
TEX_Gustatory	3.891*** (0.273)
TEX_Olfactory	−3.132*** (0.405)
IMA_Visual	−0.516** (0.206)
IMA_Auditory	1.730*** (0.633)
IMA_Tactile	0.768*** (0.199)
IMA_Gustatory	−0.647* (0.374)
IMA_Olfactory	1.555*** (0.346)
TEX_IMA_Disparity	−1.277*** (0.160)
TEX_Length	0.0000307** (0.0000137)
TEX_Readability	0.0125*** (0.00237)
IMA_H	0.00265*** (0.000315)
IMA_S	−0.00216*** (0.000257)
IMA_V	−0.00260*** (0.000264)
Time fixed effect	Yes
Hotel fixed effect	Yes
Observations	7333
Log-Likelihood	−17030.534
Pseudo R2	0.5319

The usefulness of reviews is mainly evident in consumer voting, where potential customers indicate their appreciation for reviews they find helpful (Luo et al., 2021). However, this behavior is generally influenced by the sensory stimulation of the content, leading to evaluations of usefulness. The decision-making process of potential consumers is often influenced by the multisensory elements in content such as text and images, thereby affecting their final decisions (Li et al., 2022). The impact of sensory elements in multimodal online reviews have recently garnered increasing attention (Brochado et al., 2021). Even so, few are aware of the sensory effects present in textual and image content, or the discrepancies between text and images. We aim to explore the multisensory elements (i.e., visual, auditory, tactile, olfactory, gustatory) in multimodal online review content, focusing on the perceived value for potential consumers. Multimodal LLMs and Computer Vision (CV) technology are employed to extract sensory elements (i.e., text, images) from multimodal online reviews. Econometrics and machine learning models are employed to investigate the relationships between multisensory elements and review usefulness, thereby addressing existing knowledge gaps. The conclusions and discussion are presented below.

First, there is a significant positive correlation between the auditory, tactile, and gustatory elements in the text content of multimodal online reviews and the perceived usefulness of reviews. This correlation can be attributed to the significant role of potential consumers' psychological expectations. This observation concurs with existing research (Oliver, 1980; Sun et al., 2024). When potential consumers read online reviews, the text content recreates the authentic experiences of actual consumers through detailed descriptions and directly influences their perception of the review's value by stimulating sensory elements such as auditory, tactile, and gustatory cues, thereby affecting their preconceived cognitive frameworks. Similarly, there is a significant positive correlation between the auditory, tactile, and olfactory elements in the image content of multimodal online reviews and the perceived usefulness of the reviews. This correlation further explains the important role of potential consumers' psychological expectations from another perspective. Images can stimulate the sensory cognition of potential consumers through their intuitive content, thereby influencing their perception of the review's value. Additionally, sensory marketing theory also suggests that consumer decisions are influenced not only by rational analysis but also by sensory stimuli (Krishna, 2012; Schmitt, 1999). These stimuli can trigger consumers' psychological expectations, thereby enhancing the perceived value of reviews. In multimodal online reviews, the auditory, tactile, and gustatory elements in the text, as well as the auditory, tactile, and olfactory elements in the images enhance the credibility and perceived usefulness of the reviews by stimulating consumers' sensory experiences. This phenomenon validates the role of sensory integration (i.e., the synergistic effect of multisensory stimuli) (Elder & Krishna, 2010; Mattila & Wirtz, 2001) and emotional connection (i.e., multisensory stimuli triggering consumer experiential responses) (Mehraliyev et al., 2020) in sensory marketing theory, demonstrating that multisensory stimuli significantly influence the enhancement of review value.

Second, it is noteworthy that this study further reveals a significant negative correlation between the discrepancies in multisensory elements in both the text and image content of multimodal online reviews and the perceived usefulness of the reviews. This phenomenon is attributed to

the fact that when the information conveyed by text and images is inconsistent, potential consumers experience cognitive conflict, which reduces their trust in the perceived value of the review (Mattila & Wirtz, 2001; Veloso et al., 2024; Zhang et al., 2023). Several studies (Chatterjee, 2020; Lee & Choi, 2021; Li et al., 2022; Li, Kwok, et al., 2023; Manthiou et al., 2020; Sykora et al., 2022) have confirmed from an emotional perspective that inconsistencies between text and images can lead to changes in consumer attitudes. For example, Li et al. (2022) examined the impact of discrepancies in emotional expression between review text and images on review usefulness and review enjoyment, highlighting the critical role of text-image emotional discrepancies in shaping consumer behavior. Additionally, Lee and Choi (2021) explored the impact of inconsistencies between images and text in product descriptions on consumer cognition. They found that when discrepancies exist between the image and text, consumers typically use the text as the reference point for interpreting the information. If the value conveyed by the image conflicts with that described in the text, consumers' negative emotions and cognitive reactions intensify. This study further expands on this theory by highlighting that multisensory discrepancies are a key factor influencing the perceived usefulness of online reviews. These findings extend the research on multisensory cues from text-based cues to visual perceptual cues, contributing to a deeper understanding of the important role of perceptual psychology and sensory marketing in multimodal online reviews (Fan et al., 2023; Krishna, 2012; Mandler, 2014; Peck & Childers, 2008; Petit et al., 2019)

Third, this study also found a significant negative correlation between the visual elements in the text and image content and the perceived usefulness of the reviews. First, from a cognitive psychology perspective, the text content stimulates potential consumers' cognitive memory, especially visual cognition, through detailed descriptions, which in turn triggers memories of other sensory elements (Herz & Engen, 1996). Meanwhile, when potential consumers are exposed to image content, the visual elements first make an impact, followed by the activation of other sensory elements in their cognition. This has been confirmed by Gibson (1979). This suggests that visual elements play a guiding role in multimodal reviews, triggering sensory memories and assisting potential consumers in cognitive processing across other sensory dimensions. However, when the stimulation of these sensory elements primarily relies on visual information, it can impose significant cognitive pressure on potential consumers, thereby weakening their trust in the perceived value of the review. According to cognitive load theory (Sweller, 1988), the complexity of information processing depends on the individual's allocation of cognitive resources in a specific context. When confronted with the visual elements in text and image content, potential consumers' cognitive load is often high, as visual stimuli, being the primary avenue for information input, quickly occupy their cognitive resources. This high cognitive load can lead to a counterproductive effect in processing the review content, thereby impacting the effective perception and evaluation of the information (Sweller, 1988). Therefore, when potential consumers are exposed to multimodal sensory information from text and images, the excessive cognitive load can result in counterproductive effects, a finding confirmed by our study. This finding not only advances the research on cognitive load theory in relation to multisensory elements in multimodal content, but also deepens the understanding of the role of sensory cognition and memory in potential consumers.

5.2. Theoretical contributions

The research presents several significant theoretical contributions. First, existing literature on multisensory elements in online reviews has primarily focused on textual cues (e.g., Elder & Krishna, 2010; Mehraliyev et al., 2020), largely overlooking the multisensory aspects embedded in review photos. Our study is among the earliest to systematically examine the presence of multisensory elements in both text and images of online reviews and their impact on review usefulness. Specifically, we not only identified the differential effects of multisensory elements in review text on usefulness but also uncovered the unique role of multisensory elements in review images in influencing perceived usefulness. This advances the understanding of multimodal online reviews, particularly their persuasion effect on tourist decision-making in the tourism and hospitality industry. Furthermore, existing research on multimodal online reviews has primarily focused on quality assessment and sentiment analysis (e.g., Choi & Leon, 2020; Li et al., 2022; Sun et al., 2024). For instance, Li et al. (2022) examined the impact of text-image sentiment discrepancy in multimodal online reviews on review usefulness. Building on sensory marketing theory, we extend the research perspective in this field. When examining the influence of multimodal online review content on consumers' perceived review usefulness, multisensory cues should also be taken into account.

Second, this study introduces an automated method for identifying multisensory elements by integrating machine learning models with Multimodal LLMs, thereby offering a novel methodological framework for sensory research. Existing studies have largely relied on manual scoring or experimental methods to evaluate sensory elements in social media data, which are susceptible to subjective bias, inefficiency, and constraints imposed by experimental conditions (e.g., Helmefalk & Hultén, 2017; Krishna et al., 2010; Li et al., 2019). Theoretical, we redefine sensory elements as quantifiable and modelable features, moving beyond traditional subjective assessments. This framework elucidates the attributes of individual sensory elements and their representation in multimodal data, laying the groundwork for a more systematic and objective evaluation of sensory elements. Additionally, our method provides a standardized analytical tool for sensory research, addressing issues of inconsistency and lack of generalizability in existing studies, thereby advancing the development of sensory marketing theory.

Third, this study makes an important contribution to the existing sensory marketing literature by exploring the impact of sensory discrepancy in multimodal online reviews. Existing research has primarily focused on the interactions between different sensory modalities, such as vision and auditory (Russell, 2002), often neglecting the discrepancies in sensory elements across various data modalities. For instance, auditory elements conveyed through text may differ significantly from those depicted in images. By integrating sensory marketing theory (Krishna, 2012; Schmitt, 1999), cognitive load theory (Sweller, 1988) and introducing the concept of sensory discrepancies, our study highlights the value of the discrepancy between text and image sensory elements.

5.3. Practical implications

Based on the findings of this study, several practical contributions can be made to enhance the effectiveness of multimodal online review, particularly within the hospitality industry. By strategically managing sensory elements in online reviews, hotels can optimize the value of these reviews for both current and potential customers, thereby strengthening their market position (Verhoef et al., 2009).

First, our work provides hotel marketers with an avenue to gain deeper insights into how consumers perceive and acquire multisensory experiences from multimodal online review content. These sensory details create a more immersive and life-like experience for potential consumers, enabling them to conduct a multidimensional evaluation of their future stay, ultimately supporting more informed decision-making. Scholars have suggested that high-quality multisensory experiences embedded in online reviews can significantly enhance subsequent hotel sales (Sun et al., 2025). However, it is important to note that excessive or irrelevant sensory cues may negatively impact review usefulness. Therefore, hotel managers should selectively encourage customers to provide meaningful and relevant sensory descriptions by offering rewards such as discounts, loyalty points, or complimentary services. These incentives can motivate customers to share richer, more

comprehensive reviews while ensuring that the sensory details enhance, rather than detract from, the overall usefulness of the reviews. Additionally, such reviews can gain greater visibility on social media platforms, thereby attracting a broader audience.

Second, hotel marketers should closely monitor sensory cues in multimodal online reviews that negatively impact perceived review usefulness. Our findings suggest that certain sensory descriptions may reduce the perceived value of reviews, thereby limiting their effectiveness in assisting potential consumers. By identifying and addressing these negative sensory cues, hotel practitioners can enhance the overall quality and usefulness of online reviews, ultimately supporting consumers in making more informed decisions. Additionally, our research findings highlight the importance of consistency across different sensory modalities. Inconsistencies in sensory cues may lead to consumer confusion or distrust, thereby undermining the effectiveness of online reviews. Therefore, hotel practitioners should focus on ensuring that multimodal online reviews maintain coherence in sensory descriptions to enhance their value for potential customers. By addressing these issues, hotels can improve the quality of multimodal online reviews, which in turn supports consumers in making more accurate and informed decisions. Ultimately, this contributes to the credibility and effectiveness of the e-commerce platform as a whole.

Third, this study highlights the importance of enhancing the analytical capabilities of online review platforms by integrating advanced technologies. Specifically, incorporating computer vision and natural language processing can enable platforms to more effectively analyze and interpret sensory cues embedded in both text and images. Online review platforms can leverage our proposed multisensory cue quantification method to develop an extensible multimodal content analysis framework for online review and electronic commerce websites. The proposed multisensory cue quantification method will systematically assess the impact of multisensory cues in reviews on consumers' perceived usefulness, offering actionable insights into how sensory descriptions shape consumer decision-making. By implementing this analysis framework, practitioners can not only identify high-value reviews but also optimize recommendation algorithms by prioritizing reviews with impactful sensory cues. Such optimization enhances recommendation relevance, improves user experience, and drives higher platform engagement. In addition, this technological integration would allow platforms to identify positive sensory descriptions, ensuring a more accurate depiction of the sensory attributes of a product or service. In turn, this would enhance the overall usefulness and reliability of reviews, providing more valuable insights for potential consumers.

Last but not least, the innovative AI method proposed in this study, designed to measure consumer multisensory experience from multimodal online reviews, offers hotel marketers and managers a transformative tool for real-time monitoring of consumer experiences. This capability is of paramount importance because multisensory experiences can influence the perceived usefulness of reviews, which, as established in prior research, significantly influences consumer decision-making and, ultimately, hotel sales (Sun et al., 2025).

5.4. Limitations and future research

Several limitations apply to this research and should be investigated in the future. First, regarding the extraction of sensory elements from text reviews, we have constructed an English Dictionary by identifying keywords from the text content and expanding them using the *Word2vec* method. Future studies could further validate the accuracy of sensory element extraction from texts, such as Multimodal LLMs (Dogru et al., 2023; Wong et al., 2023). Second, in the exploration of sensory elements within images, we employed a large language model for image-to-text conversion, which has shown strong effectiveness in understanding sensory elements. However, we encourage future studies to explore direct extraction of sensory elements from images, rather than relying on conversion to text. Alternative methods, such as advanced image analysis techniques, could be investigated to further enhance the accuracy and reliability of sensory element identification (Ren et al., 2021). Third, the data used in our study were limited to a single city (i.e., Hong Kong, China) and a specific type of hotels (i.e., five-star). Subsequent research could test the generalizability of our findings by employing online review data from other types of hotels beyond Hong Kong.

CRediT authorship contribution statement

Shaolong Sun: Writing – review & editing, Funding acquisition, Conceptualization. **Haoqiang Sun:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Haozhe Xu:** Writing – review & editing, Writing – original draft, Software, Methodology. **Hengyun Li:** Supervision, Project administration, Investigation. **Shouyang Wang:** Writing – review & editing, Project administration, Investigation.

Impact statement

This study highlights the crucial role of multisensory cues in consumer perception, significantly enhancing the practical value of multimodal online reviews in the hospitality industry. By encouraging rich sensory descriptions in online reviews, hotels can attract potential customers and build stronger trust. Offering incentives for detailed reviews and enhancing the sensory appeal of services can increase the hotel's appeal on social platforms, fostering customer loyalty and engagement. For hotels, leveraging diverse sensory cues between text and images reinforces brand image and customer satisfaction. The integration of advanced technologies like multimodal large language model and natural language processing in social platforms represents a significant innovation, providing richer sensory information and improving service quality. This technological advancement elevates market competitiveness in the tourism sector, enhances customer satisfaction, and promotes overall service excellence, contributing positively to the broader social and economic landscape.

Declaration of generative AI and AI-assisted technologies in the writing process. Statement

During the preparation of this work the authors used ChatGPT 4o to check grammar and polish writing. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that there is no conflict of interests regarding the publication of this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tourman.2025.105206.

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